SMOOTHNESS IS INHERITED BY THE SLICE CUT OUT BY A FIBREWISE NONZERODIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10590-10609 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10611-10658, the complete author proof (46 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Smooth morphisms and the differential criterion for smoothness, nonzerodivisors on fibres, and quotients by a regular element; the computation is retained.

Source reference. Original lemma algebra-lemma-one-equation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10658–10658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma divisors-lemma-relative-Cartier: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10658–10658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-slice-given-element: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6166–6177. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-differentials-relative-immersion: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10658–10658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-dimension-fibre-at-a-point: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10658–10658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-smooth-at-point: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10658–10658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ be a point with image $s \in S$. Let $h \in \mathfrak{m}_x \subset \mathcal{O}_{X,x}$. Assume*

(1) *f is smooth at x, and*
(2) *the image $d\overline{h}$ of dh in*

$$\Omega_{X_s/s,x} \otimes_{\mathcal{O}_{X_s,x}} \kappa(x) = \Omega_{X/S,x} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$$

is nonzero.

Then there exists an affine open neighbourhood $U \subset X$ of x such that h comes from $h \in \Gamma(U, \mathcal{O}_U)$ and such that $D = V(h)$ is an effective Cartier divisor in U with $x \in D$ and $D \to S$ smooth.

Proof. As f is smooth at x we may assume, after replacing X by an open neighbourhood of x that f is smooth. In particular we see that f is flat and locally of finite presentation. By Lemma lemma-slice-given-element we already know there exists an open neighbourhood $U \subset X$ of x such that h comes from $h \in \Gamma(U, \mathcal{O}_U)$ and such that $D = V(h)$ is an effective Cartier divisor in U with $x \in D$ and $D \to S$ flat and of finite presentation. By Morphisms, Lemma morphisms-lemma-differentials-relative-immersion we have a short exact sequence

$$\mathcal{C}_{D/U} \to i^*\Omega_{U/S} \to \Omega_{D/S} \to 0$$

where $i : D \to U$ is the closed immersion and $\mathcal{C}_{D/U}$ is the conormal sheaf of D in U. As D is an effective Cartier divisor cut out by $h \in \Gamma(U, \mathcal{O}_U)$ we see that $\mathcal{C}_{D/U} = h \cdot \mathcal{O}_S$. Since $U \to S$ is smooth the sheaf $\Omega_{U/S}$ is finite locally free, hence its pullback $i^*\Omega_{U/S}$ is finite locally free also. The first arrow of the sequence maps the free generator h to the section $\mathrm{d}h|_D$ of $i^*\Omega_{U/S}$ which has nonzero value in the fibre $\Omega_{U/S,x} \otimes \kappa(x)$ by assumption. By right exactness of $\otimes\kappa(x)$ we conclude that

$$\dim_{\kappa(x)} \big(\Omega_{D/S,x} \otimes \kappa(x)\big) = \dim_{\kappa(x)} \big(\Omega_{U/S,x} \otimes \kappa(x)\big) - 1.$$

By Morphisms, Lemma morphisms-lemma-smooth-at-point we see that $\Omega_{U/S,x} \otimes \kappa(x)$ can be generated by at most $\dim_x(U_s)$ elements. By the displayed formula we see that $\Omega_{D/S,x} \otimes \kappa(x)$ can be generated by at most $\dim_x(U_s) - 1$ elements. Note that $\dim_x(D_s) = \dim_x(U_s) - 1$ for example because $\dim(\mathcal{O}_{D_s,x}) = \dim(\mathcal{O}_{U_s,x}) - 1$ by Algebra, Lemma algebra-lemma-one-equation (also $D_s \subset U_s$ is effective Cartier, see Divisors, Lemma divisors-lemma-relative-Cartier) and then using Morphisms, Lemma morphisms-lemma-dimension-fibre-at-a-point. Thus we conclude that $\Omega_{D/S,x} \otimes \kappa(x)$ can be generated by at most $\dim_x(D_s)$ elements and we conclude that $D \to S$ is smooth at x by Morphisms, Lemma morphisms-lemma-smooth-at-point again. After shrinking U we get that $D \to S$ is smooth and we win. □